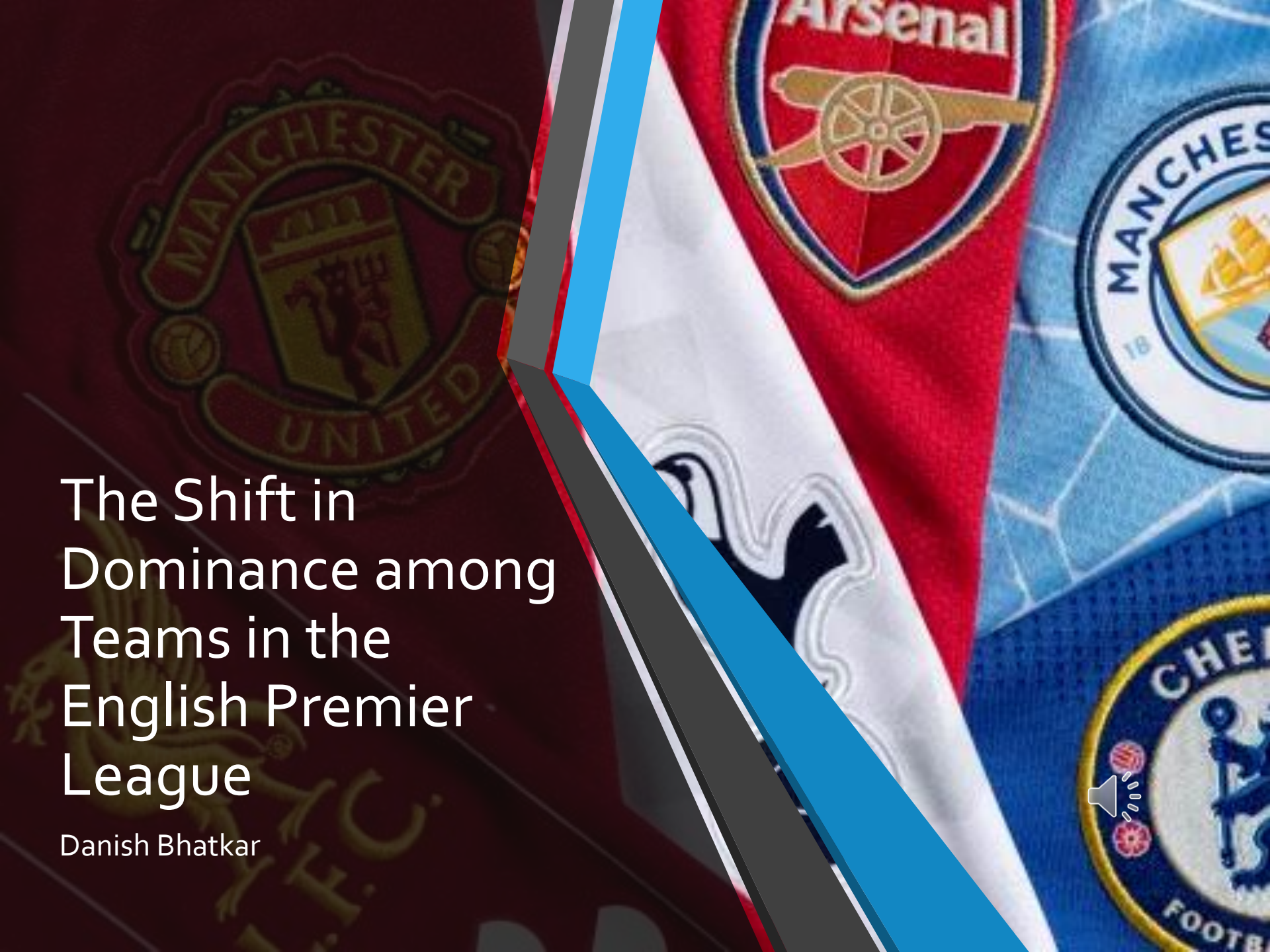




The Shift in Dominance among Teams in the English Premier League

Danish Bhatkar



About English Premier League

- Elite professional league in England – formed in 1992
- Consists of 20 teams – each team plays 38 matches in a season
- Every season the bottom 3 teams are relegated and the top 3 teams from the Championship division (second-tier) are promoted to Premier League
- Some historic clubs are Manchester United, Chelsea, Arsenal, Liverpool and Manchester City
- Known for fast-paced and physically demanding style of play
- Considered as one of the best Football Leagues across the globe



Source Data : Premier League Matches 1993-2023

- The source data (shown on the right) was obtained from [Kaggle](https://www.kaggle.com/datasets/evangower/premier-league-matches-19922022)
- Each row in the data provides details about all the matches played in the last 30 years

Variable	Definition
Season_Year_End	Year in which the season ends (A season starts in August and ends in May in the following year)
Wk	Week number of the season
Date	Matchday Date
Home	Home Team
HomeGoals	Goals Scored by Home Team
AwayGoals	Goals Scored by Away Team
Away	Away Team
FTR	H – Home Team won D – Draw A – Away Team won

Season_End_Year	Wk	Date	Home	HomeGoals	AwayGoals	Away	FTR
1993	1	8/15/1992	Coventry City	2	1	Middlesbrough	H
1993	1	8/15/1992	Leeds United	2	1	Wimbledon	H
1993	1	8/15/1992	Sheffield Utd	2	1	Manchester Utd	H
1993	1	8/15/1992	Crystal Palace	3	3	Blackburn	D
1993	1	8/15/1992	Arsenal	2	4	Norwich City	A
1993	1	8/15/1992	Ipswich Town	1	1	Aston Villa	D
1993	1	8/15/1992	Everton	1	1	Sheffield Weds	D
1993	1	8/15/1992	Southampton	0	0	Tottenham	D
1993	1	8/15/1992	Chelsea	1	1	Oldham Athletic	D
1993	1	8/16/1992	Nott'ham Forest	1	0	Liverpool	H
1993	1	8/17/1992	Manchester City	1	1	QPR	D
1993	2	8/18/1992	Blackburn	1	0	Arsenal	H
1993	2	8/18/1992	Wimbledon	0	1	Ipswich Town	A
1993	2	8/19/1992	Tottenham	0	2	Coventry City	A
1993	2	8/19/1992	Liverpool	2	1	Sheffield Utd	H
1993	2	8/19/1992	Aston Villa	1	1	Leeds United	D
1993	2	8/19/1992	Oldham Athletic	1	1	Crystal Palace	D
1993	2	8/19/1992	Middlesbrough	2	0	Manchester City	H
1993	2	8/19/1992	Norwich City	2	1	Chelsea	H
1993	2	8/19/1992	QPR	3	1	Southampton	H
1993	2	8/19/1992	Manchester Utd	0	3	Everton	A
1993	2	8/19/1992	Sheffield Weds	2	0	Nott'ham Forest	H
1993	3	8/22/1992	Tottenham	2	2	Crystal Palace	D
1993	3	8/22/1992	Wimbledon	1	2	Coventry City	A
1993	3	8/22/1992	Blackburn	1	0	Manchester City	H
1993	3	8/22/1992	Norwich City	1	1	Everton	D
1993	3	8/22/1992	Sheffield Weds	3	3	Chelsea	D
1993	3	8/22/1992	Manchester Utd	1	1	Ipswich Town	D
1993	3	8/22/1992	Aston Villa	1	1	Southampton	D
1993	3	8/22/1992	Middlesbrough	4	1	Leeds United	H
1993	3	8/22/1992	QPR	3	2	Sheffield Utd	H
1993	3	8/22/1992	Oldham Athletic	5	3	Nott'ham Forest	H
1993	3	8/23/1992	Liverpool	0	2	Arsenal	A
1993	4	8/24/1992	Southampton	0	1	Manchester Utd	A
1993	4	8/25/1992	Sheffield Utd	2	2	Wimbledon	D

Data Source -

<https://www.kaggle.com/datasets/evangower/premier-league-matches-19922022>

Source Data : Premier League Matches 1993-2023

The dataset contains information about every match played in the Premier League from 1993 to 2023. The original purpose of the data was to analyze the following:-

- Best performing team in the last 30 years
- Best way to collect points, attacking or defensive style
- Goals scored and conceded by various teams
- Head-to-head record
- Advantage of playing on the home turf

Accessing, Exploring and Cleaning of Source Data

- Source data was available in **csv** format
- PROC IMPORT step implemented to **access** and store data in SAS
- PROC CONTENTS step implemented to **analyze** data attributes
 - There are 3 char variables and 5 integer variables as seen on the right
- Compressed 'Home', 'Away' and 'FTR' columns to ensure there are no white spaces to avoid duplication and **clean** the data
- PROC MEANS step for understanding **base level statistics** of all numeric variables

```
/* Importing raw data */
proc import datafile="/home/dbhatka/EPG294/Final Project/Data Files/premier-league-matches.csv"
    dbms=csv out=rawdata replace;
    guessingrows=max;
run;

/* Cleaning dataset and saving it to a folder for future use */

data finproj.cleandata (rename=(Season_End_Year = YearEnd));
    set rawdata;
    Home = compbl(Home);
    Away = compbl(Away);
    FTR = strip(FTR);
run;
```

```
proc contents data=finproj.cleandata; run;
```

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat
7	Away	Char	15	\$15.	\$15.
6	AwayGoals	Num	8	BEST12.	BEST32.
3	Date	Num	8	YYMMDD10.	YYMMDD10.
8	FTR	Char	1	\$1.	\$1.
4	Home	Char	15	\$15.	\$15.
5	HomeGoals	Num	8	BEST12.	BEST32.
2	Wk	Num	8	BEST12.	BEST32.
1	YearEnd	Num	8	BEST12.	BEST32.

Variable	N	Mean	Std Dev	Minimum	Maximum
YearEnd	12026	2007.71	9.0725590	1993.00	2023.00
Wk	12026	19.7305006	11.1239157	1.0000000	42.0000000
Date	12026	17428.49	3319.24	11915.00	23158.00
HomeGoals	12026	1.5243639	1.3064166	0	9.0000000
AwayGoals	12026	1.1425245	1.1339300	0	9.0000000

Accessing, Exploring and Cleaning of Source Data

Snapshot of the cleaned dataset

YearEnd	Wk	Date	Home	HomeGoals	AwayGoals	Away	FTR
1993	1	1992-08-15	Coventry City	2	1	Middlesbrough	H
1993	1	1992-08-15	Leeds United	2	1	Wimbledon	H
1993	1	1992-08-15	Sheffield Utd	2	1	Manchester Utd	H
1993	1	1992-08-15	Crystal Palace	3	3	Blackburn	D
1993	1	1992-08-15	Arsenal	2	4	Norwich City	A
1993	1	1992-08-15	Ipswich Town	1	1	Aston Villa	D
1993	1	1992-08-15	Everton	1	1	Sheffield Weds	D
1993	1	1992-08-15	Southampton	0	0	Tottenham	D
1993	1	1992-08-15	Chelsea	1	1	Oldham Athletic	D
1993	1	1992-08-16	Nott'ham Forest	1	0	Liverpool	H
1993	1	1992-08-17	Manchester City	1	1	QPR	D
1993	2	1992-08-18	Blackburn	1	0	Arsenal	H
1993	2	1992-08-18	Wimbledon	0	1	Ipswich Town	A
1993	2	1992-08-19	Tottenham	0	2	Coventry City	A
1993	2	1992-08-19	Liverpool	2	1	Sheffield Utd	H

Variable	Definition
YearEnd	Year in which the season ends (A season starts in August and ends in May in the following year)
Wk	Week number of the season
Date	Matchday Date
Home	Home Team
HomeGoals	Goals Scored by Home Team
AwayGoals	Goals Scored by Away Team
Away	Away Team
FTR	H – Home Team won D – Draw A – Away Team won

Preparing Data

Following tables were prepared for analysis purposes:-

- SortedData – A combined league table sorted in the ascending order of 'YearEnd' and descending order of the points collected
- Ranking – Ranks added to the SortedData table as per the points collected in each season.
- Winners – Displays only the title winners in the last 30 years along with various attributes like goal difference, total points, etc.
- PointsTable – Provides the total number of points scored by each team in all seasons
- PointsTable (by decade) – Decade column added to the SortedData

Please note:

Several temporary tables were created that were used in preparing the above permanent tables. The reason for explaining only the permanent tables is because only these tables were used for all the analysis procedures

Sorted Data

Year End	Team	Goals For	Goals Against	Goal Diff	Total Points
1993	Manchester Utd	67	31	36	84
1993	Aston Villa	57	40	17	74
1993	Norwich City	61	65	-4	72
1993	Blackburn	68	46	22	71
1993	QPR	63	55	8	63
1993	Liverpool	62	55	7	59
1993	Sheffield Weds	55	51	4	59
1993	Tottenham	60	66	-6	59
1993	Manchester City	56	51	5	57
1993	Arsenal	40	38	2	56
1993	Chelsea	51	54	-3	56
1993	Wimbledon	56	55	1	54
1993	Everton	53	55	-2	53
1993	Coventry City	52	57	-5	52
1993	Ipswich Town	50	55	-5	52

Winners

Year End	Team	Goals For	Goals Against	Goal Diff	Total Points
1993	Manchester Utd	67	31	36	84
1994	Manchester Utd	80	38	42	92
1995	Blackburn	80	39	41	89
1996	Manchester Utd	73	35	38	82
1997	Manchester Utd	76	44	32	75
1998	Arsenal	68	33	35	78
1999	Manchester Utd	80	37	43	79
2000	Manchester Utd	97	45	52	91
2001	Manchester Utd	79	31	48	80
2002	Arsenal	79	36	43	87
2003	Manchester Utd	74	34	40	83
2004	Arsenal	73	26	47	90
2005	Chelsea	72	15	57	95
2006	Chelsea	72	22	50	91
2007	Manchester Utd	83	27	56	89

Total Points (All Seasons)

Team	Total Points
Manchester Utd	2441
Arsenal	2225
Chelsea	2182
Liverpool	2176
Tottenham	1847
Manchester City	1718
Everton	1610
Newcastle Utd	1481
Aston Villa	1419
West Ham	1298
Southampton	1088
Blackburn	970
Leicester City	821
Leeds United	820
Fulham	692

Rankings

Year End	Team	Goals For	Goals Against	Goal Diff	Total Points	Rank
1993	Manchester Utd	67	31	36	84	1
1993	Aston Villa	57	40	17	74	2
1993	Norwich City	61	65	-4	72	3
1993	Blackburn	68	46	22	71	4
1993	QPR	63	55	8	63	5
1993	Liverpool	62	55	7	59	6
1993	Sheffield Weds	55	51	4	59	7
1993	Tottenham	60	66	-6	59	8
1993	Manchester City	56	51	5	57	9
1993	Arsenal	40	38	2	56	10
1993	Chelsea	51	54	-3	56	11
1993	Wimbledon	56	55	1	54	12
1993	Everton	53	55	-2	53	13
1993	Coventry City	52	57	-5	52	14
1993	Ipswich Town	50	55	-5	52	15

Points Table (by decade)

Year End	Team	Goals For	Goals Against	Goal Diff	Total Points	Decade
1993	Manchester Utd	67	31	36	84	1993-2003
1994	Manchester Utd	80	38	42	92	1993-2003
1995	Blackburn	80	39	41	89	1993-2003
1996	Manchester Utd	73	35	38	82	1993-2003
1997	Manchester Utd	76	44	32	75	1993-2003
1998	Arsenal	68	33	35	78	1993-2003
1999	Manchester Utd	80	37	43	79	1993-2003
2000	Manchester Utd	97	45	52	91	1993-2003
2001	Manchester Utd	79	31	48	80	1993-2003
2002	Arsenal	79	36	43	87	1993-2003
2003	Manchester Utd	74	34	40	83	1993-2003
2004	Arsenal	73	26	47	90	2003-2013
2005	Chelsea	72	15	57	95	2003-2013
2006	Chelsea	72	22	50	91	2003-2013
2007	Manchester Utd	83	27	56	89	2003-2013

Terminology for Prepared Data Tables

Variable	Definition
YearEnd	Year in which the season ends (A season starts in August and ends in May in the following year)
Team	Team name
GoalsFor	Goals scored by the corresponding team
GoalsAgainst	Goals conceded by the corresponding team
GoalDiff	Difference between goals scored and goals conceded
TotPoints / TotalPoints	Total points scored by the corresponding team
Rank	Rank of a team in the corresponding season (YearEnd)
Decade	Decade to which the row pertains

Data Analysis

Analyzing the Performances of All Teams

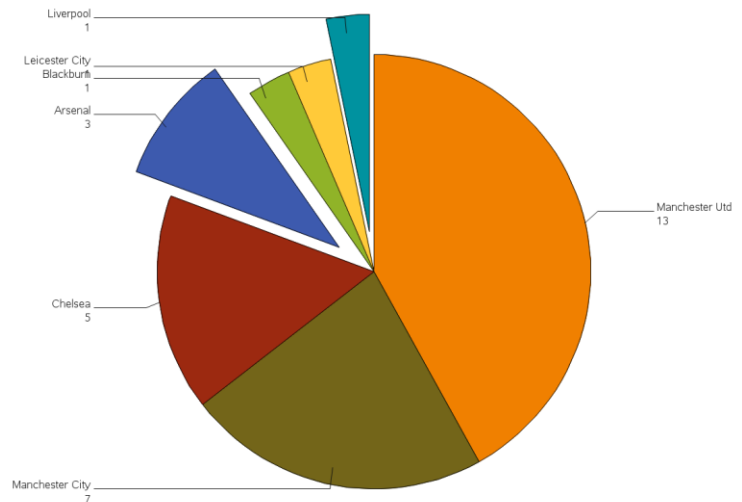
- First pie chart indicates the number of titles won by each team (top 7)
- Second pie chart displays the total points scored in all the seasons by each team (top 7)
- Arsenal and Liverpool have very few titles but are in the top 4 considering the points scored
- Everton and Tottenham are among the top 7 teams with regards to the overall points scored but have not won any titles in the 30-year-period

Since Manchester United, Manchester City, Arsenal, Chelsea and Liverpool are the common 5 clubs in both the visuals, we can consider them as the top 5 clubs in the last 30 years

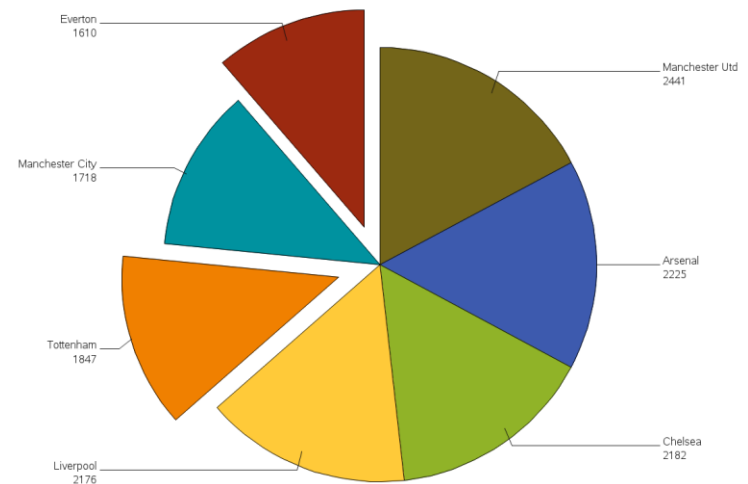
```
title "Number of Titles won from 1993 to 2023";
proc gchart data=finproj.winners;
  pie Team / descending clockwise other=2 value=arrow explode='Liverpool' 'Arsenal' noheading;
run;
quit;
title;

title "Total Number of Points Scored from 1993 to 2023";
proc gchart data=finproj.pointsTotal;
  pie Team / sumvar=TotalPoints descending clockwise other=2 value=arrow
    explode='Tottenham' 'Everton' noheading;
  where TotalPoints > 1600;
run;
quit;
title;
```

Number of Titles won from 1993 to 2023



Total Number of Points Scored from 1993 to 2023

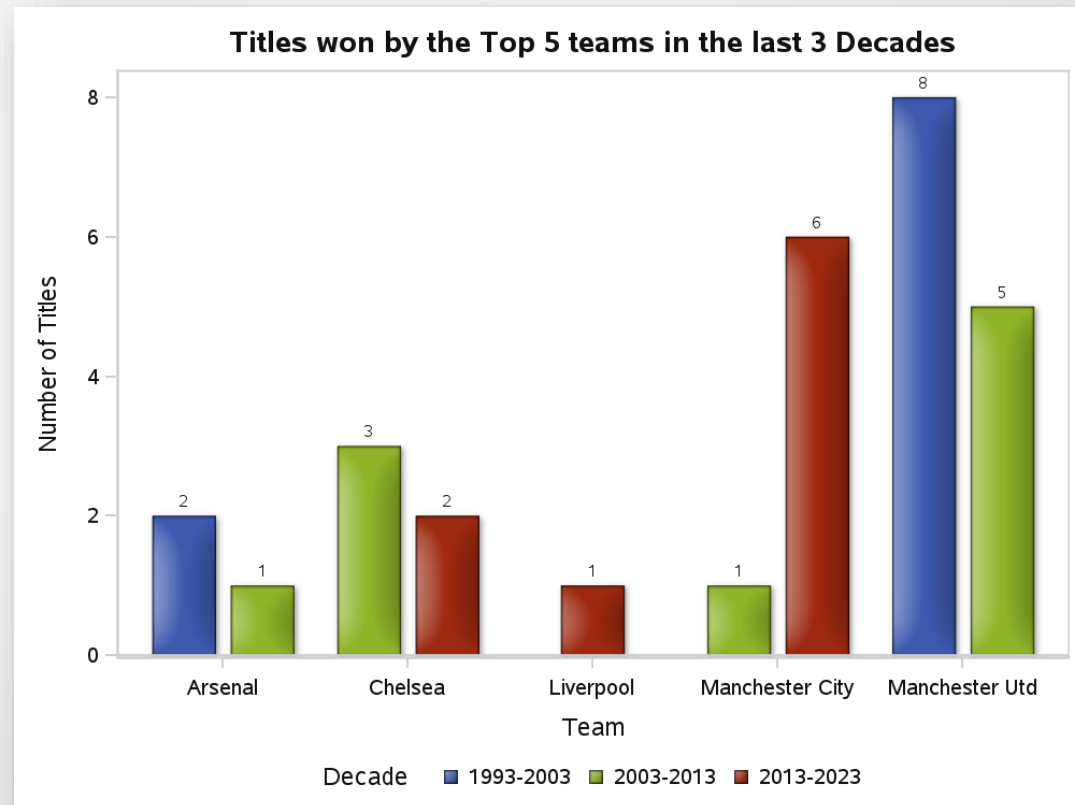


Data Analysis

Performance of Top 5 Teams by Decade

- The clustered bar graph displays the titles won by the top 5 teams in each decade.
- All teams except Manchester City have a decreasing trend
- Manchester United and Manchester City top the charts followed by Chelsea
- Manchester United has not won a single title in the last decade

For further analysis, we can consider Manchester United, Manchester City (looking at their domination), and Chelsea (since the club has won 5 titles and won 2 of them in the last decade itself).



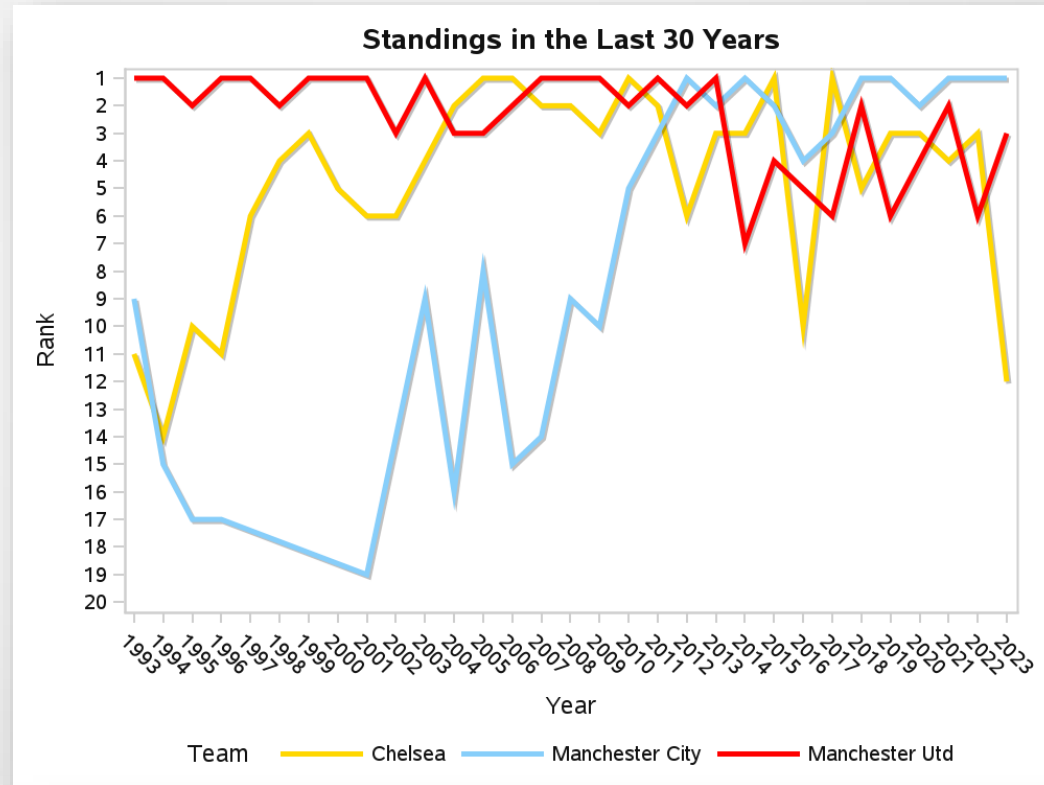
```
title "Titles won by the Top 5 teams in the last 3 Decades";  
proc sgplot data=finproj.decadesorting;  
  vbar Team / barwidth=0.8 dataskin=matte group=Decade groupdisplay=cluster grouporder=data datalabel;  
  label YearEnd = 'Number of Titles';  
  yaxis label='Number of Titles';  
  where Team in ('Manchester Utd' 'Manchester City' 'Arsenal' 'Chelsea' 'Liverpool');  
run;  
title;
```

Data Analysis

Analyzing the performance trends

- The line chart shows the ranks of the top 3 teams in the last 30 years
- Chelsea has been quite inconsistent in the last decade
- Manchester United has lost their edge in recent years
- Meanwhile, Manchester City has a steep improvement in its performance

Chelsea was topping the charts for a short interval of time. On the other hand, a shift of power can be seen from Manchester United to Manchester City.



```
title "Standings in the Last 30 Years";
proc sgplot data=finproj.ranking;
  vline YearEnd / response=Rank stat=mean group=Team dataskin=crisp;
  styleattrs datacontrastcolors=(gold lightskyblue red);
  yaxis values=(1 to 20) reverse;
  where Team in ('Manchester Utd' 'Manchester City' 'Chelsea');
  Label Rank = Rank YearEnd = Year;
run;
title;
```

Data Analysis

Analyzing the Attacking/Defensive strategies

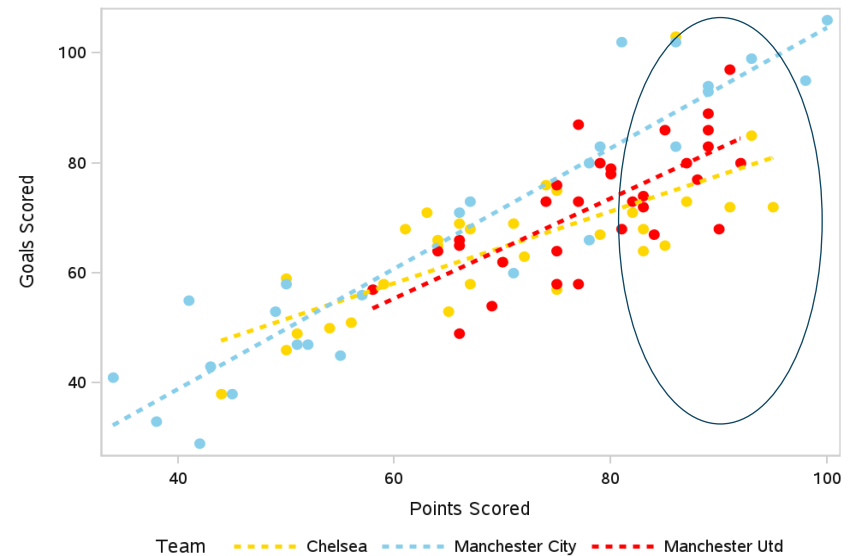
- The first scatter plot displays the number of goals scored against points scored in each season and the second chart displays the number of goals conceded against points scored in each season
- Both the charts have a median line to visualize the trend
- Focus must be only on the right end of both the charts since this area explains the attacking and defensive strategies of the teams when they are at their best (most points scored)

Manchester City is the best at goalscoring among the top 3 teams whereas Chelsea is quite well at defending goals when at its best. It can be inferred that Manchester City is reliant on their attackers whereas Chelsea depends on its defenders to perform well.

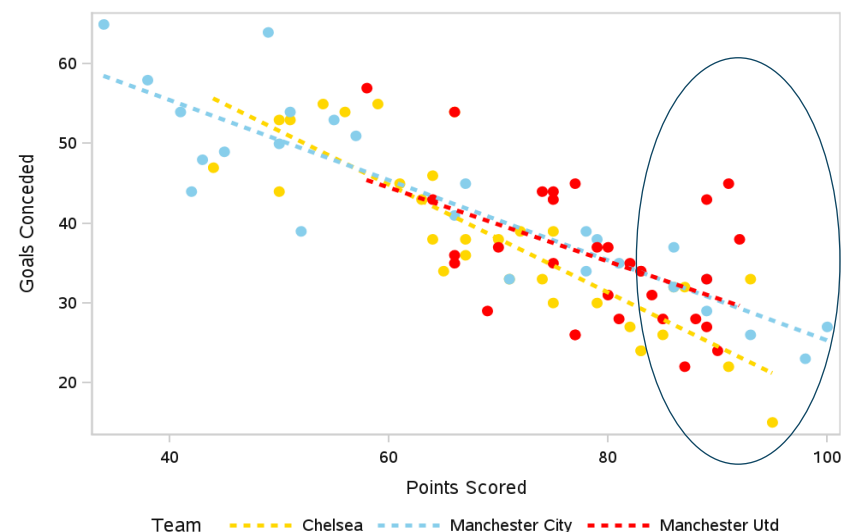
```
proc sgplot data=finproj.ranking;
  reg x=TotPoints y=GoalsFor / group=Team;
  where Team in ('Manchester Utd' 'Manchester City' 'Chelsea');
  styleattrs datacontrastcolors=(red skyblue gold) datalinepatterns=(2);
  label GoalsFor = 'Goals Scored' TotPoints = 'Points Scored';
  title 'Comparison of Goals Scored against Points Scored for the Top 3 Teams';
run;
title;

proc sgplot data=finproj.ranking;
  reg x=TotPoints y=GoalsAgainst / group=Team;
  where Team in ('Manchester Utd' 'Manchester City' 'Chelsea');
  styleattrs datacontrastcolors=(red skyblue gold) datalinepatterns=(2);
  label GoalsAgainst = 'Goals Conceded' TotPoints = 'Points Scored';
  title 'Comparison of Goals Conceded against Points Scored for the Top 3 Teams';
run;
title;
```

Comparison of Goals Scored against Points Scored for the Top 3 Teams



Comparison of Goals Conceded against Points Scored for the Top 3 Teams



Predictive Analysis

Correlation of Total Points against Scoring and Conceding of Goals

- Implemented PROC CORR step to calculate Pearson's and Spearman's Correlation Coefficients
- The correlation coefficient explains the relativity of Total Points with the Goals Scored and Goals Conceded by a team
- The Total Points parameter is more strongly related to Goals Scored as compared to Goals Conceded in both the calculations

It can be inferred that if Team A scores more goals than Team B, then Team A is likely to score more points than Team B.

Pearson Correlation Coefficients, N = 626

	GoalsFor	GoalsAgainst
TotPoints Total Points	0.88579	-0.81598

Spearman Correlation Coefficients, N = 626

	GoalsFor	GoalsAgainst
TotPoints Total Points	0.86412	-0.81146

```
title 'Correlation of Total Points against Scoring and Conceding of Goals';
proc corr data=finproj.ranking PEARSON spearman noprob;
  var GoalsFor GoalsAgainst;
  with TotPoints;
  label TotPoints='Total Points' GoalsFor = 'Goals Scored' GoalsAgainst = 'Goals Conceded';
run;
title;
```

Simple Statistics

Variable	N	Mean	Std Dev	Median	Minimum	Maximum	Label
TotPoints	626	52.68530	16.34248	50.00000	11.00000	100.00000	Total Points
GoalsFor	626	51.23323	14.99717	48.00000	20.00000	106.00000	Goals Scored
GoalsAgainst	626	51.23323	12.77748	51.00000	15.00000	100.00000	Goals Conceded

Predictive Analysis

Linear Regression Method

Predicting Total Number of Points Required by a Team to Win the Premier League
(Based on the last 15 years to conduct analysis as per the recent trend)

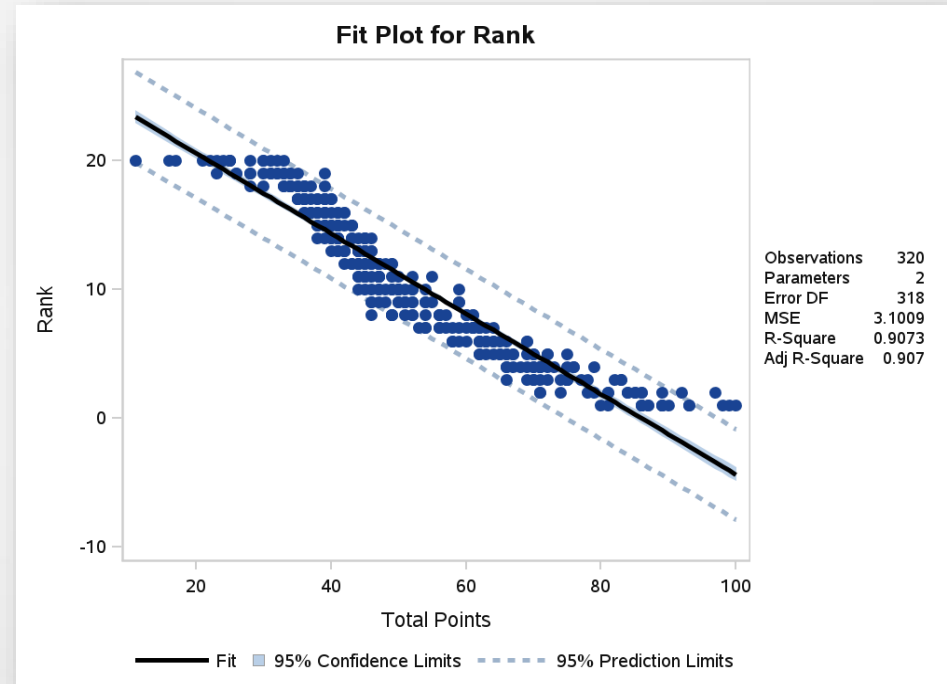
Number of Observations Read	320
Number of Observations Used	320

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	9653.92959	9653.92959	3113.32	<.0001
Error	318	986.07041	3.10085		
Corrected Total	319	10640			

Root MSE	1.76092	R-Square	0.9073
Dependent Mean	10.50000	Adj R-Sq	0.9070
Coeff Var	16.77070		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	26.82235	0.30865	86.90	<.0001
TotPoints	Total Points	1	-0.31177	0.00559	-55.80	<.0001

```
proc reg data=finproj.ranking plots(only) = fitplot;  
  model Rank = TotPoints;  
  where YearEnd >= 2008 and YearEnd <=2023;  
  label TotPoints = 'Total Points';  
  title 'Regression Analysis';  
  title2 'Predicting Total Number of Points Required by a Team to Win the Premier League';  
  title3 '(Based on the last 15 years to conduct analysis as per the recent trend)';  
run;  
title; title1; title2;
```



- PROC REG method implemented to process regression analysis by considering the “Rank” as dependent variable on “TotalPoints” variable.
- Linear Regression provides a linear relation between multiple variables. The equation in this case can be read as :-

$$\text{Rank} = 26.822 - (0.312 * \text{TotalPoints})$$

- For Rank=1, we get TotalPoints = 82.825. Thus, we can assume that a team scoring 83 points or more has high chances of winning a title

Summarizing the Results

Based on the above analysis, we can make the following observations:-

- Clearly, Manchester United is the most successful team in the last 30 years winning 13 titles and scoring the most points in all seasons. (slide 11)
- Although, there are teams such as Arsenal, Liverpool, and Tottenham who have been consistent in scoring points, they have won a total of 4 titles only amongst themselves. (slide 11)
- Chelsea did dominate the league between 2004 and 2011 but since then the journey for the team has been downhill. (slide 13)
- Among the top 3 teams, only Chelsea and Manchester City have won titles in the last decade, that hints at the downfall of Manchester United. (slide 12)
- Manchester United and Manchester City have opposite slopes of progress as seen above. (slide 13)
- Manchester City's strength is scoring goals while Chelsea's strength is its defense. Manchester United stands on the weaker side in both the cases. (slide 14)

Considering the above points, it can be concluded that Manchester City has been the best team in recent years and the trajectory has been upwards.

The dominance of English Football has shifted from Manchester United to Manchester City.

Exporting Data

- After the completion of data table cleaning, modifications and analysis procedures, all the steps were grouped together
- ODS PowerPoint method implemented to output the entire data to a pptx file
- Snow style added in the options to provide an aesthetic look to the charts
- Exported pptx file was modified further by providing explanatory data

```
options nodate;
ods powerpoint file="/home/dbhatka/EPG294/Final Project/FinalProject.pptx" style=snow;

proc contents data=finproj.cleandata; run;
proc means data=finproj.cleandata; run;

title 'Clean Data';
proc print data=finproj.cleandata (obs=15); run;
title 'Sorted Data';
proc print data=finproj.sortedtable (obs=15); run;
title 'Rankings';
proc print data=finproj.ranking (obs=15); run;
title 'Winners';
proc print data=finproj.winners (obs=15); run;
title 'Points Table';
proc print data=finproj.pointsTotal (obs=15); run;
title 'Points Table by Decade';
proc print data=finproj.decadeSorting (obs=15); run;

ods graphics on;
title "Number of Titles won from 1993 to 2023";
proc gchart data=finproj.winners;
title;

title "Total Number of Points Scored from 1993 to 2023";
proc gchart data=finproj.pointsTotal;
title;

title "Titles won by the Top 5 teams in the last 3 Decades";
proc sgplot data=finproj.decadesorting;
title;

title "Standings in the Last 30 Years";
proc sgplot data=finproj.ranking;
title;

proc sgplot data=finproj.ranking;
title;
proc sgplot data=finproj.ranking;
title;

title 'Correlation of Total Points against Scoring and Conceding of Goals';
proc corr data=finproj.ranking PEARSON spearman noprob;
var GoalsFor GoalsAgainst;
with TotPoints;
label TotPoints='Total Points' GoalsFor = 'Goals Scored' GoalsAgainst = 'Goals Conceded';
run;
title;

proc reg data=finproj.ranking plots(only) = fitplot;
model Rank = TotPoints;
where YearEnd >= 2008 and YearEnd <=2023;
label TotPoints = 'Total Points';
title 'Regression Analysis';
title2 'Predicting Total Number of Points Required by a Team to Win the Premier League';
title3 '(Based on the last 15 years to conduct analysis as per the recent trend)';
run;
title; title1; title2;
ods graphics off;

ods powerpoint close;
```

References

- <https://www.kaggle.com/datasets/evangower/premier-league-matches-19922022>
- https://documentation.sas.com/doc/en/pgmsascdc/9.4_3.5/pgmsaswlcmlcm/home.htm
- <https://www.90min.com/posts/the-2022-premier-league-table>

About Me

- My name is Danish Bhatkar and I'm from Mumbai, India.
- I'm a Computer Science Grad student at Clemson University, interested in the Data Science field.
- My plan is to complete my studies and graduate in May 2025.
- I've been a football fan and a Manchester United fan since 2011 and that's the reason for me choosing this dataset

